

The Future of Drug Discovery

AI That Simulates Biology

- Why drug discovery is still slow
- The vision: simulate the cell
- CZ Biohub and the data infrastructure
- What is Perturb-seq?
- Stack and Perturb Sapiens
- X-Cell (Xaira)
- Protein language models: ESM2
- Live code demo

The \$2 Billion Problem: Why Drug Discovery Is Still Slow

10-15

years

average time to approval

\$2B

average cost

per approved drug

90%

failure rate

compounds that fail in trials

~20

new drugs

approved per year globally

The measurement bottleneck

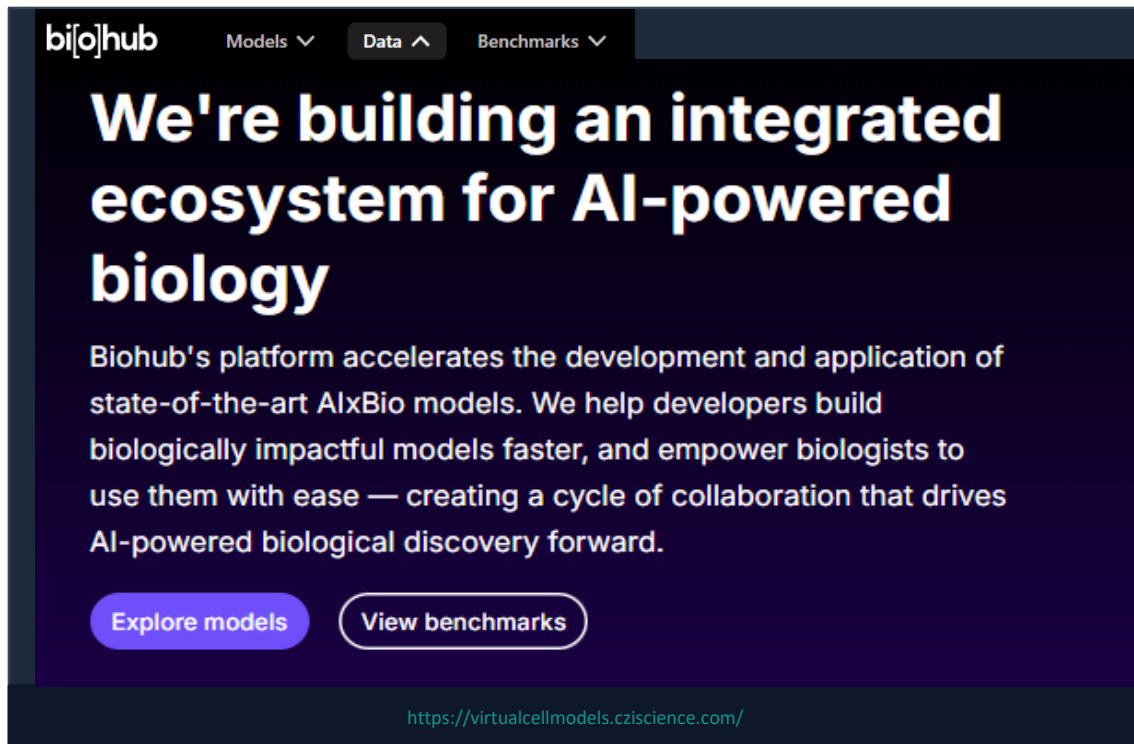
Every biological question requires a physical experiment. Test a drug in lung cancer cells? Run the assay. Does it work the same way in liver cells? Different assay. Does it hold up in primary patient cells? Another experiment, months of work.

Even with automation, you can only test so many compounds in so many contexts. The experiment space is astronomically large. Most of it has never been touched.

What if you could simulate instead of measure?

A model that has learned how cells respond to perturbations could answer biological questions in seconds instead of months. Give this drug to lung tumor cells -- what happens? What about liver cells? What about the same drug at half the dose?

This is not science fiction. Three groups published models in the last six months that begin to do exactly this. We will look at all three today.



The screenshot shows the top portion of the Biohub website. At the top is a dark navigation bar with the 'bi[o]hub' logo on the left and three menu items: 'Models' with a downward arrow, 'Data' with an upward arrow, and 'Benchmarks' with a downward arrow. Below the navigation bar is a large dark blue section with white text. The main headline reads 'We're building an integrated ecosystem for AI-powered biology'. Below this is a paragraph: 'Biohub's platform accelerates the development and application of state-of-the-art AIxBio models. We help developers build biologically impactful models faster, and empower biologists to use them with ease — creating a cycle of collaboration that drives AI-powered biological discovery forward.' At the bottom of this section are two rounded buttons: a blue one labeled 'Explore models' and a white one with a blue border labeled 'View benchmarks'. At the very bottom of the screenshot is a dark footer bar containing the URL 'https://virtualcellmodels.cziscience.com/' in green text.

bi[o]hub Models ▾ Data ▲ Benchmarks ▾

We're building an integrated ecosystem for AI-powered biology

Biohub's platform accelerates the development and application of state-of-the-art AIxBio models. We help developers build biologically impactful models faster, and empower biologists to use them with ease — creating a cycle of collaboration that drives AI-powered biological discovery forward.

[Explore models](#) [View benchmarks](#)

<https://virtualcellmodels.cziscience.com/>

\$500M commitment (April 2026)

The Virtual Biology Initiative: five-year program to build predictive models of life. \$100M to coordinate global data generation, \$400M for data at scale and next-generation measurement technology.

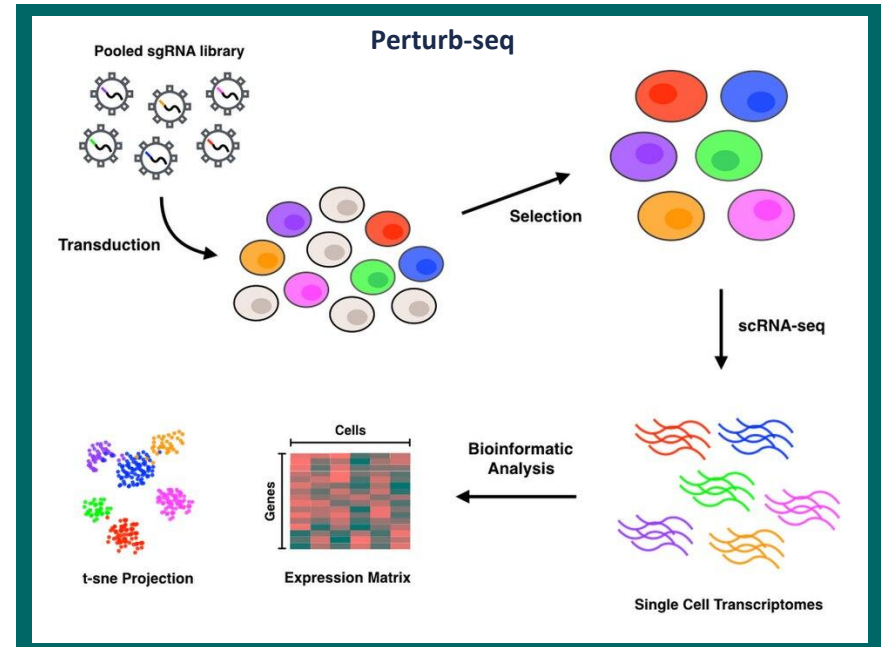
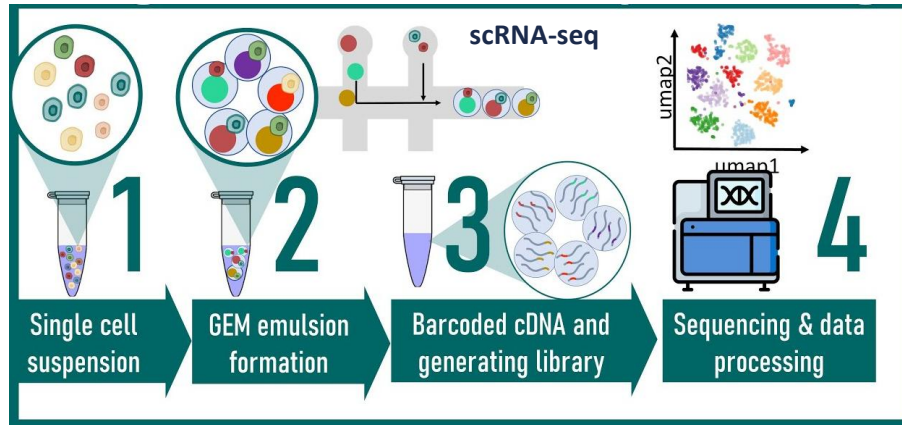
The Billion Cells Project

17 institutions including MIT, Stanford, UCSF, Columbia, ETH Zurich. Coordinates the largest ever single-cell sequencing effort to break the data bottleneck limiting biological AI models.

The goal

Create the technologies and multi-modal datasets needed to build predictive models of life. For the first time, it is becoming possible to envision a high-accuracy predictive model of the cell.

The Training Data: Single-Cell RNA Sequencing and Perturb-seq



scRNA-seq

Measure the gene expression of thousands of individual cells simultaneously. Each cell gets a barcode. The output is a matrix: cells x genes, with expression levels for all 20,000+ human genes per cell.

Perturb-seq

Use CRISPR to knock out one gene at a time, then immediately measure gene expression with scRNA-seq. Repeat for thousands of genes. The result: a map of cause and effect in the cell.

Why this matters for AI

This is causal data. Not just what cells look like, but what happens when you change something. This is the training signal that makes virtual cell models possible.

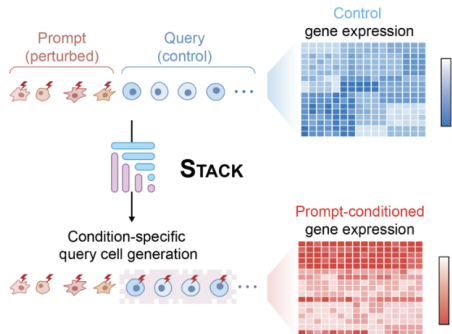
Stack: In-Context Learning for Single-Cell Biology (Arc Institute, January 2026)

JANUARY 9, 2026

arcinstitute.org/news/foundation-model-stack

STACK: Simulating cellular conditions via prompt engineering, without the need for fine-tuning

Arc Institute unveils a new foundation model with in-context learning of single-cell biology, and applies it to generate Perturb Sapiens, an atlas of simulated human cells



Perturb Sapiens: A Human Whole-Organism Atlas of Perturbed Cells

Dataset Description

huggingface.co/datasets/arcinstitute/Perturb-Sapiens

Total file size:

1.34 TB

Perturb Sapiens is an evolving database of AI-predicted single-cell perturbation responses, representing the **first human whole-organism atlas of perturbed cells**.

Metric	Value
Total Cells	103+ million
Human Tissues	28
Cell Classes	40
Drug Conditions	111 (control: Dimethyl sulfoxide)
Cytokine Conditions	90 (control: PBS)

What Stack does

A foundation model trained on 149 million human single cells. Key innovation: in-context learning. Like a large language model answering questions without fine-tuning, Stack predicts cellular responses in new contexts it was never trained on -- just by using real cells as prompts.

Perturb Sapiens atlas

The output of applying Stack at scale. Predicted drug and cytokine responses across 28 human tissues, 40 cell types, and 203 perturbations. Available on HuggingFace as h5ad files today.

112 drugs including EGFR inhibitors. 91 cytokines. Free to download.

The clinical implication

Most drug-tissue-cell type combinations have never been experimentally measured. Doing even a fraction would cost millions of dollars and years of work. Perturb Sapiens provides predicted responses for all of them -- in seconds per query.

X-Cell: Predicting Gene Expression at Scale (Xaira Therapeutics, March 2026)

CAN BIOLOGY MOVE INTO THE MATRIX?

'Virtual cells' could transform biomedical research, but scientists are still struggling with how to model life's complexity. **By Michael Eisenstein**

As every gamer knows, computers can plausibly simulate just about anything from the routine concerns of a household to the crises confronting a multiplanetary civilization. Simulating the fundamental unit of life – the cell – should be a walk in the park. But it's not.

Each cell is a complex ecosystem of biomolecules that interact with one another and react to external cues in ways that remain poorly understood. And what's true of one cell type isn't necessarily true of another. But there is an order to the chaos.

"The cell is a complex system, and a highly robust and resilient system," says Emma Lundberg, a bioengineer at Stanford University in California. "But it's also a highly structured system – the cell has an architecture."

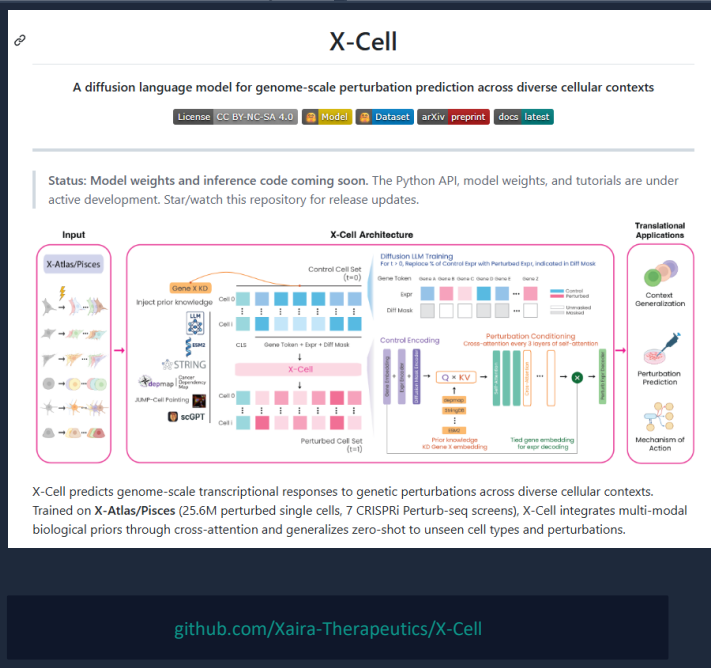
Over the past few years, researchers have begun reverse-engineering that architecture into 'virtual cells' – models that

recreate the internal environment of cells by when responding to external triggers.

Several teams are now tapping the power of transcriptomic (genomic) and other data sets to build models that reveal the underlying biological logic and possible angles for the intervention. "We have to think about this as a means of getting towards a target and for me, that goal is to be able to iterate the hypothesis search process."

Roohani, a machine-learning researcher at the ARC Institute in Palo Alto, California. The field remains far short of a

functional virtual cell, however. "I don't think people would sensibly want to claim that they have built a virtual cell unless they need to



Published March 2026

Trained on 25.6 million perturbed single-cell transcriptomes across 16 biological contexts. Up to 4.9 billion parameters.

The key result

Predicted T cell activation gene expression changes even though it had never been trained on that process. Zero-shot generalization to a new biological context.

Status: frontier

The pip package installs. The model architecture is defined. Full inference weights are being finalized -- watch the repository. This is the cutting edge: the paper is out, the code is being released in real time.

Protein Language Models: Learning the Language of Life

Large Language Model (GPT, Claude...)

Trained on billions of words of human text.

Learns grammar, meaning, context, reasoning.

Input: text tokens | Output: text tokens

Application: write, summarize, translate, reason

Protein Language Model (ESM2...)

Trained on 250M protein sequences from all life on Earth.

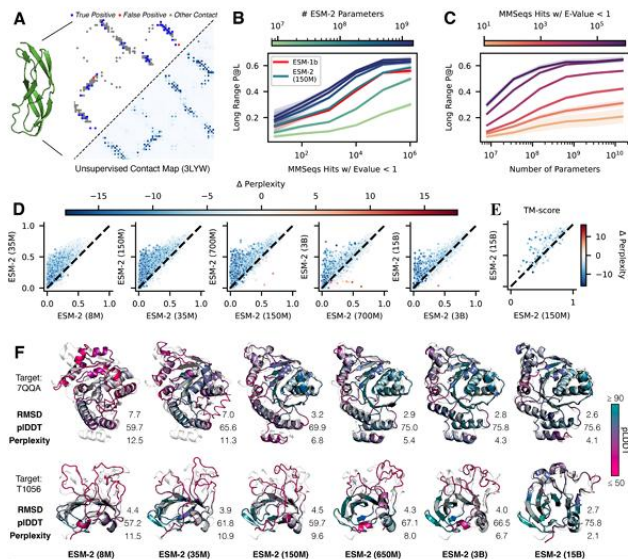
Learns amino acid grammar, structure, function, evolution.

Input: amino acid tokens | Output: embeddings

Application: predict structure, function, mutation effects

same
architecture
different
data

Evolutionary-scale prediction of atomic level protein structure with a language model



What ESM2 actually learned

By predicting masked amino acids from context (like BERT for proteins), ESM2 learned representations that encode structure, function, and evolutionary relationships -- without ever being explicitly trained on any of them.

Why this matters for drug discovery

You can embed a protein sequence and instantly see how similar it is to known drug targets, how a mutation shifts its properties, or whether a designed sequence is likely to fold and function correctly.

ESM2: 250 Million Protein Sequences, 33 Layers, 650 Million Parameters

Hugging Face

facebook / **esm2_t33_650M_UR50D** like 80 Follow AI at Meta 13.1k

ESM-2

ESM-2 is a state-of-the-art protein model trained on a masked language modelling objective. It is suitable for fine-tuning on a wide range of tasks that take protein sequences as input. For detailed information on the model architecture and training data, please refer to the [accompanying paper](#). You may also be interested in some demo notebooks ([PyTorch](#), [TensorFlow](#)) which demonstrate how to fine-tune ESM-2 models on your tasks of interest.

Several ESM-2 checkpoints are available in the Hub with varying sizes. Larger sizes generally have somewhat better accuracy, but require much more memory and time to train:

Checkpoint name	Num layers	Num parameters
esm2_t48_15B_UR50D	48	15B
esm2_t36_3B_UR50D	36	3B
esm2_t33_650M_UR50D	33	650M
esm2_t30_150M_UR50D	30	150M
esm2_t12_35M_UR50D	12	35M
esm2_t6_8M_UR50D	6	8M

250M

protein sequences in training data

33

transformer layers in the 650M model

1280

dimensions in each residue embedding

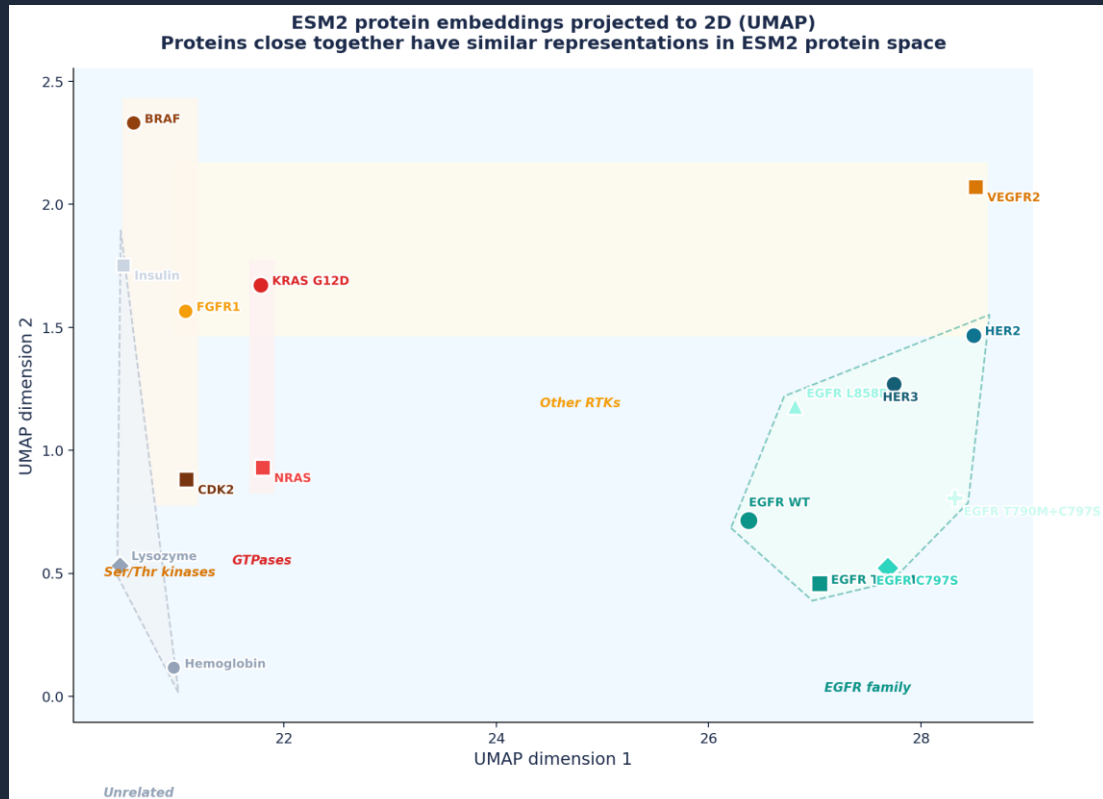
6.8B

proteins in ESM Atlas (predicted structures)

Questions we can answer with embeddings -- what we will demonstrate in the notebook

- How similar is EGFR wild-type to the T790M resistance mutant in the model's learned representation of protein space?
- Does the L858R activating mutation (which makes EGFR-driven cancers sensitive to drugs) shift the protein differently than the T790M resistance mutation?
- How does EGFR compare to KRAS and HER2 -- related but distinct oncoproteins -- in embedding space?

What We Will Build: Mapping EGFR Mutations in Protein Embedding Space



1

Load ESM2 locally

651M parameters, runs on your RTX 4000 Ada. No internet needed after initial download.

2

Embed protein panel

EGFR variants + KRAS + HER2 + random. Each becomes a 1280-dimensional vector.

3

Similarity heatmap

Pairwise cosine similarity. See which proteins the model considers similar.

4

UMAP visualization

Project 1280 dimensions to 2D. Clusters reveal model's understanding of protein space.

5

Mutation distance plot

Bar chart: how far does each EGFR mutation shift the protein from wild-type?

Where This Is Going

- From predicting genetic knockouts to predicting drug effects -- the same models, applied to compound perturbations instead of gene knockouts
- From cancer cell lines to primary patient cells -- personalized virtual cell models for individual tumor biopsies
- From single cells to tissue -- spatial transcriptomics will add location information to gene expression, enabling virtual tissue models
- From virtual cells to virtual clinical trials -- simulating patient populations before Phase II begins
- Protein language models + virtual cell models working together -- design a compound with ESM, simulate its cell response with Stack or X-Cell
- The honest caveat: predictions must be validated experimentally. These models are powerful but not perfect. They accelerate discovery; they do not replace it.

github.com/bnsreenu/AI-for-Drug-Discovery

Code in this video

- ESM2 650M protein embeddings
- EGFR mutation similarity
- Pairwise heatmap
- UMAP visualization
- Mutation distance bar chart